ARAV MENDIRATTA

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PROFILE

Computer Science Master's student at the University of Freiburg focusing on Generative AI, Business Transformation using AI, and Bioinformatics. Experienced in building cloud-native Generative AI platforms, enterprise AI value propositions, and machine learning pipelines for biological and financial data.

EXPERIENCE

Collaborative Research Center 1597 "Small Data" (University of Freiburg) **May 2026 – July 2026**
Student Researcher *Freiburg, Germany*

- Engineered an end-to-end LLM-powered bioinformatics pipeline converting high-dimensional gene expression data into structured representations (Cell2Sentence), improving rare cell-type classification accuracy by 35%.
- Designed an interactive prompt-prototyping and consensus evaluation framework to reduce batch inference error rates by 20%.
- Configured automated MLOps and GitHub Actions CI/CD workflows on high-performance computing (SLURM HPC) clusters, shortening iteration cycles by 40%.

Capgemini Germany **June 2025 – December 2025**
Data Scientist (Working Student / Werkstudent) *Germany*

Internal I&D Portfolio Development (Business Transformation & Life Sciences)

- Structured a multi-million Euro Innovation & Development (I&D) portfolio, accelerating the delivery of client-facing AI offerings for Go-To-Market (GTM) strategies by 25%.
- Executed data benchmarking via Python and Power Query, generating strategic insights that strengthened client engagement in the Life Sciences sector.
- Translated over 15 technical AI, GenAI, and Data Governance offerings into clear business value propositions to support C-level decision-making.

AUDI — Generative AI Platform

- Co-developed a Generative AI image creation platform, shortening the visual prototyping process for the AUDI design team from weeks to days.
- Designed cloud-native generative AI workflows using AWS CDK, Amazon S3, and Docker, driving a 40% efficiency increase in image generation.
- Engineered FastAPI backends handling thousands of daily requests with over 95% test coverage (Pytest) and GitHub Actions CI/CD.

EDUCATION

Albert-Ludwigs-Universität Freiburg **Oct 2024 – Expected Apr 2027**
Master of Science (M.Sc.) in Computer Science *Freiburg, Germany*

- Focus: Artificial Intelligence, Deep Learning, RNA Bioinformatics, Data-Driven Life Sciences, Digital Health, Trustworthy AI.

SRM Institute of Science and Technology **Sep 2020 – Jun 2024**
Bachelor of Technology (B.Tech.) in Computer Science and Engineering *Chennai, India*

SELECTED PROJECTS

- **Clinical Intelligence Platform (GenAI & RAG):** Built a Retrieval-Augmented Generation platform using FastAPI, OpenAI, React, and Qdrant over 50,000+ clinical documents with citation verification.
- **Single-Cell Foundation Model (Bioinformatics):** Pre-trained Transformer language models on rank-ordered gene expression profiles using PyTorch, Hugging Face Transformers, and Scanpy.
- **Geometric Drug Discovery (3D Molecular AI):** Engineered an E(3) Equivariant Graph Neural Network in PyTorch and RDKit to predict molecular binding properties from 3D atomic coordinates.

- **Workforce Intelligence Platform:** Built an explainable ML decision-support service (Scikit-learn, SHAP, FastAPI, Docker) for workforce capacity and retention planning.
- **Market Data Streaming & Cloud FX Analytics:** Engineered real-time financial data quality and analytics pipelines using Kafka, PySpark, Databricks, PostgreSQL, and Prometheus.

SKILLS & LANGUAGES

- **Generative AI & Machine Learning:** LLMs, RAG, Diffusion & LoRA, PyTorch, Hugging Face Transformers, Scikit-learn, SHAP, Graph Neural Networks, Computer Vision.
- **Bioinformatics & Data Systems:** Scanpy, AnnData, Biopython, RDKit, Databricks, PySpark, Apache Kafka, PostgreSQL, Qdrant, Power Query.
- **Cloud & Engineering:** Python, SQL, TypeScript, FastAPI, AWS (CDK, S3, EC2), Docker, GitHub Actions CI/CD, Pytest.
- **Languages:** English (Fluent / C1), German (Basic / B1), Hindi (Native).